

3.1.1	Atomic Structure	
	Fundamental particles	be able to describe the properties of protons, neutrons and electrons in terms of relative charge and relative mass
		know that early models of atomic structure predicted that atoms and ions with noble gas electron arrangements should be stable
	Protons, neutrons and electrons	understand the importance of these particles in the structure of the atom and appreciate that there are various models to illustrate atomic structure
	Mass number and isotopes	be able to recall the meaning of mass number (A) and atomic (proton) number (Z)
		be able to explain the existence of isotopes
		understand the principles of a simple mass spectrometer, limited to ionisation, acceleration, deflection and detection
		know that the mass spectrometer gives accurate information about relative isotopic mass and also about the relative abundance of isotopes
		be able to interpret simple mass spectra of elements and calculate relative atomic mass from isotopic abundance, limited to mononuclear ions
		know that mass spectrometry can be used to identify elements (as used for example in planetary space probes)
		know that mass spectrometry can be used to determine relative molecular mass
	Electron arrangement	know the electron configurations of atoms and ions up to $Z = 36$ in terms of levels and sub-levels (orbitals) s, p and d
		know the meaning of the term <i>ionisation energy</i> .
		understand how ionisation energies in Period 3 (Na – Ar) and in Group 2 (Be – Ba) give evidence for electron arrangement in sub-levels and in levels